Chapter 11: Longitudinal Analysis of Engagement

1. Introduction

Longitudinal analysis examines how student engagement evolves over the course duration. By clustering trajectory patterns, we identify distinct engagement profiles: sustained high engagement, steady decline, and late-blooming trajectories.

2. Dataset & Variables

Engagement data from 30 students across 16 weeks. Variables include engagement score, forum posts, and assignment submission status.

3. Methods

Student engagement trajectories were extracted by reshaping data into student×week matrix. Trajectories were standardized and clustered using K-means (k=3). Cluster means provide prototypical engagement patterns.

4. Results

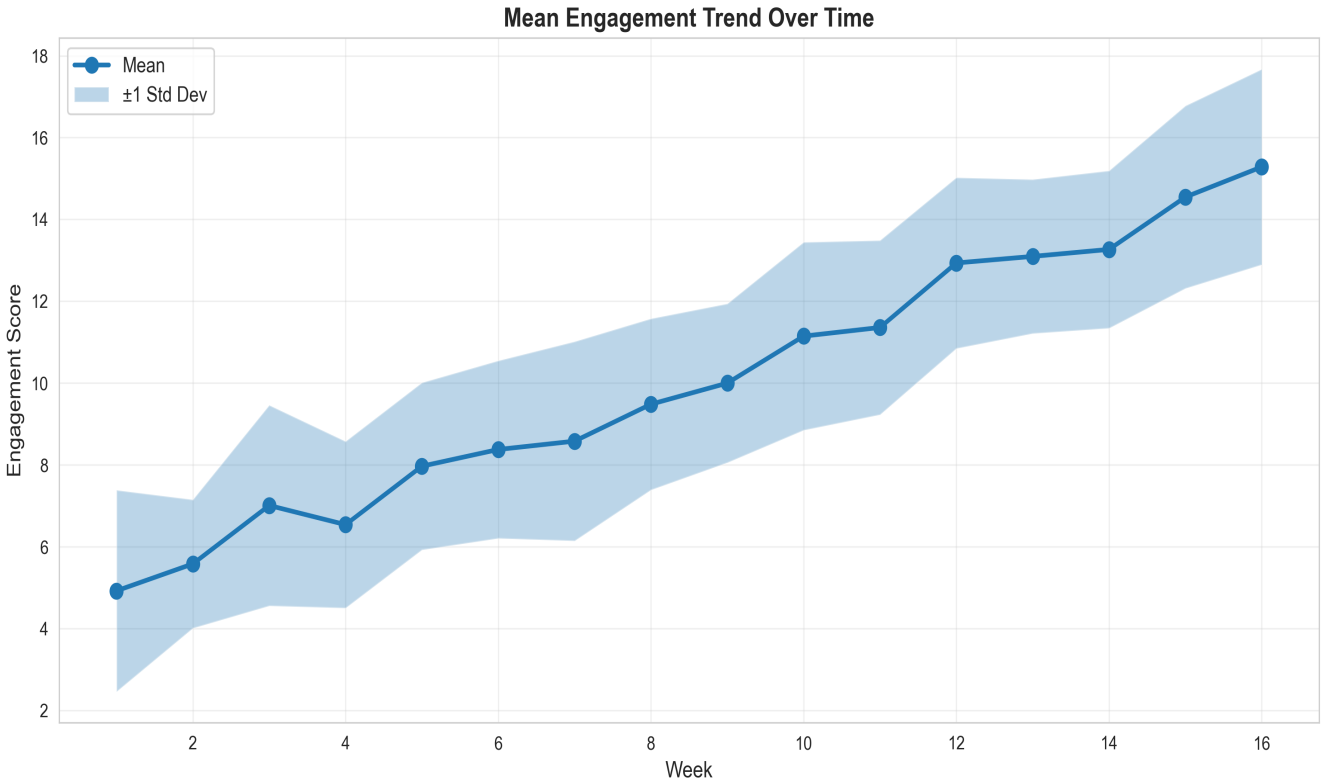

Fig 1: Overall engagement trend with confidence band.

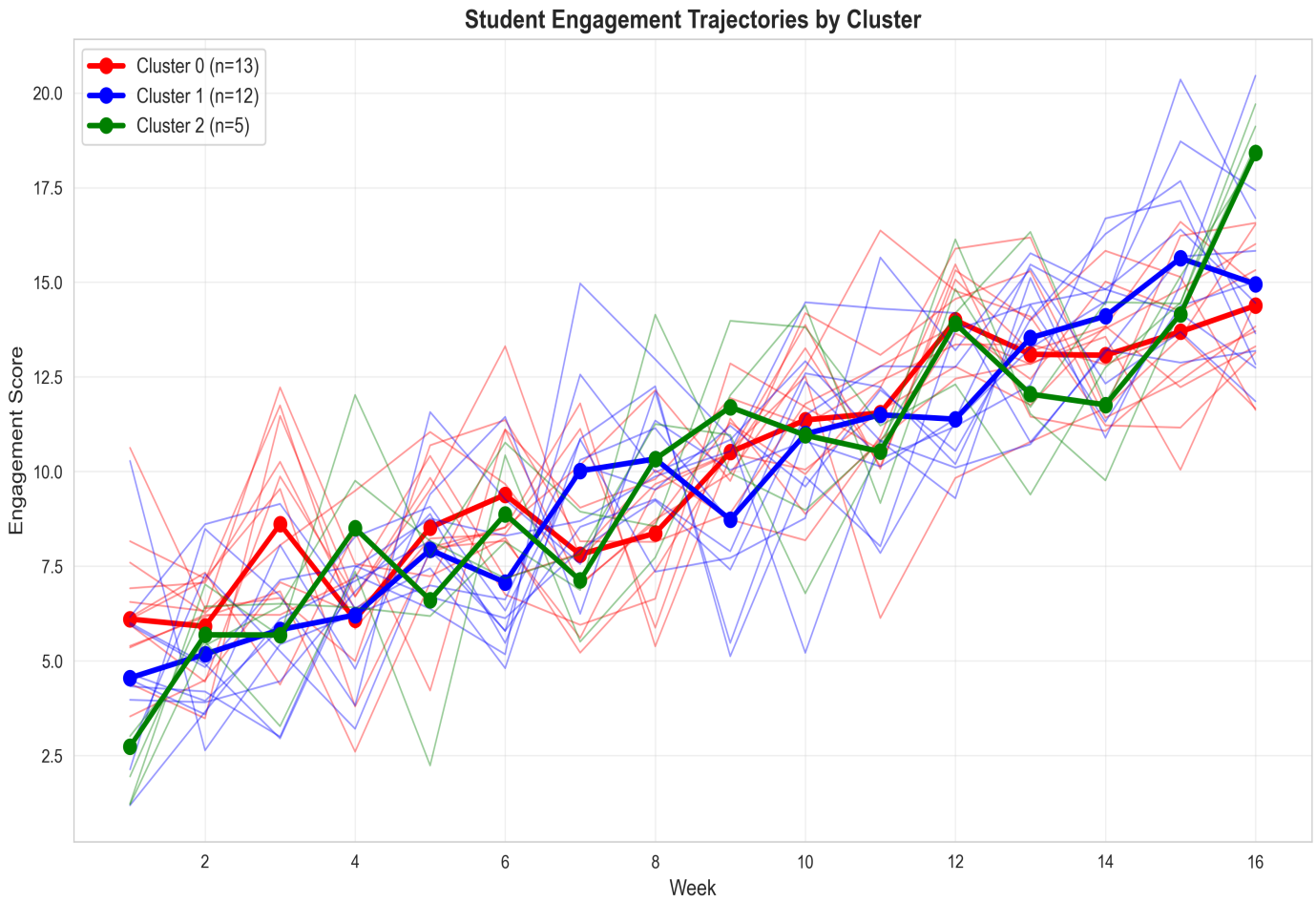

Fig 2: Three distinct trajectory clusters.

5. Interpretation

Three distinct engagement patterns emerge: sustained high engagement, moderate declining engagement, and low engagement. Identifying these patterns enables targeted interventions for at-risk learners.

6. Limitations

Engagement scores are synthetic for this demonstration. Real analyses require domain-validated metrics. Temporal autocorrelation not explicitly modeled.

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